# Mridulla Ganesh

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## Education

**Purdue University** West Lafayette, IN
*Computer Engineering* *August 2023 — December 2026*

- Cumulative GPA: 3.6/4.0 | Dean's List, Semester Honors
- Relevant Coursework: ASIC Design Lab, Microprocessor Systems and Interfacing, Data Structures in C

## Work Experience

**Undergraduate Teaching Assistant** August 2025 — Present
Purdue University *West Lafayette, IN*

- Mentoring undergraduate students in embedded systems programming in C for the RP2350 microcontroller
- Guiding students in writing programs to interface with hardware peripherals, inclduing GPIO, UART, SPI, and timers
- Performing root cause analysis on student projects with logic errors and hardware faults, teaching problem solving at the hardware-software interface

**Quality Engineering Intern** May 2025 — August 2025
Coherent, Inc. *Monroe, CT*

- Wrote Python scripts using OpenCV to collect and process image data for an AI defect detection model, helping improve the model's detection accuracy for surface flaws on semiconductor wafer chucks
- Built interactive PowerBI dashboards to visualize quality metrics, providing the engineering team with clearer insights into manufacturing trends
- Assisted with root cause analysis on a critical component failure by performing physical inspections and identifying the cause

## Research Experience

**Research with VIP SoCET, Datacenter Network Accelerator** August 2025 — Present
Purdue University *West Lafayette, IN*

- Developing a software simulation of a server environment, creating a model of a custom harware network accelerator to enable pre-silicon validation
- Implementing network communication layers to bridge the software simulation with the physical hardware accelerator over Ethernet, creating a testing environment
- Designing and scripting test workloads to validate the accelerator's functionality and benchmark its performance

**Undergraduate Research Assistant, AI Network Performance** January 2025 — Present
Purdue University, w/ Dr. Xiaoqi Chen *West Lafayette, IN*

- Analyzing modern data center network architecture to model their performance impact on large-scale AI training workloads
- Utilizing network simulators including ns-3 and AstraSim to analyze data flow, identifying key network bottlenecks in collective communication patterns

**Undergraduate Research Assistant, GPU Acceleration** January 2025 — May 2025
Purdue University, w/ Dr.Milind Kulkarni *West Lafayette, IN*

- Accelerated k-nearest neighbors algorithm by leveraging NVIDIA GPU ray-tracing cores for high-speed spatial queries
- Optimized the core search algorithm in CUDA and C++, using rotations to optimize for data in different Minowsky distances

**Research with VIP Semiconductors @ Birck, Polymer Team** January 2025 — May 2025
Purdue University *West Lafayette, IN*

- Operated a CO2 laser to pattern PET substrates, and determined optimal parameters for feature creation
- Contributed to the chip integration process by preparing and applying epoxy for backfilling
- Prepared samples for adhesion tests by spin coating an adhesive layer onto the PET substrate

## Skills

- **Programming Languages**: C, Python, Java, MATLAB, RISC-V Assembly, C++, CUDA
- **Technologies**: Linux, Git, ns-3, System Verilog, LTspice, Autodesk Eagle